Run report: endo-chain-bench counting and timing runs, 2026-10-16

`research/endo_native_bench_20261016/` · first run started 2026-10-16 17:59 UTC · commit `33e2dd10bd8b75a13122683576bf5a81f1583577`

Descriptive record of the six runs (one counting and one timing run per curve). Observations only; the interpretation is in the study's results report (`../../REPORT.pdf`) and `README.md`. Every value is copied from the committed records; the graphs are drawn from them by `../../figures/plot.py`.

1. What ran

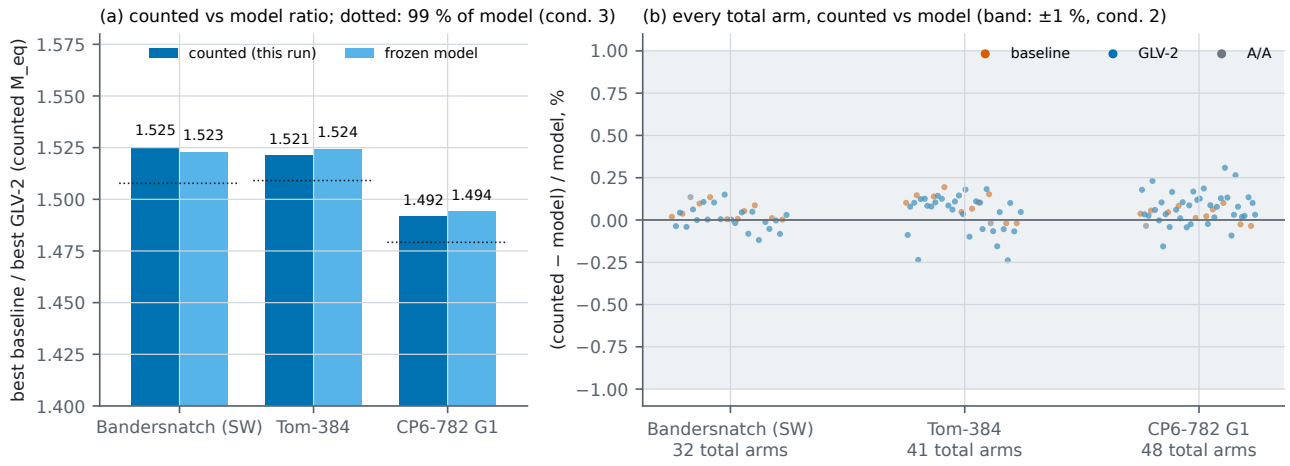
- **Command.** `endo-chain-bench` (harness `85793758e`), protocol committed in `PROTOCOL.md` at `33e2dd10b` before any run on these curves. Counting: `target/opcount/release/endo-chain-bench --count-only --chains … --expect-sha256 … --model … --reference … --pairs 256 --seed 20261016`. Timing: the same binary without the opcount feature, `--rounds 20`, under `isolated_bench run --wait --cpus 3 --settle 5 --max-other-cpu 0.5`. The full command list is in `README.md`, "Commands".
- **Build.** `rustc 1.97.0 (2d8144b78 2026-07-07)`, release, no `target-cpu`; field over `Uint<L>::mont_mul` with 4 / 6 / 13 limbs.
- **Host.** Intel(R) Xeon(R) Processor @ 2.10GHz, cloud VM, 4 logical CPUs, kernel 6.18.44-fc-v139. Isolation earned: L0 (shared VM, timing pinned to CPU 3 and admitted by `isolated_bench`; counting runs not isolated, counts do not depend on contention).
- **Workloads.** 256 (k, P) pairs per curve, seed 20261016; timing 20 rounds, arm order forward on even and reversed on odd rounds; statistic median and minimum of round wall time / pairs.
- **Arms.** Baselines (width-w NAF, Jacobian and affine tables, $w = 3..7$), GLV-2 grid on the declared chain (generic / optimised evaluator × table × width), GLV elements at `optimised/jacobian/w5`, $\varphi(P)$ and decomposition stage diagnostics, one A/A arm. 38 / 96 / 180 arms.
- **Controls.** A/A arm (the reference re-timed); positive control before the protocol: the generic harness's counts equal the CryptoPro-B sweep's committed counts on all 97 arms (`positive_control/`).

2. Runs requested and completed

curve	arms	all correct (count / time)	timing wall s (isolated)	contended	other CPU-s	invol. switches	major faults
Bandersnatch	38	true / true	23.3	false	0.48	55	0
Tom-384	96	true / true	98.9	false	1.79	136	0
CP6-782 G1	180	true / true	570.2	false	11.11	621	0

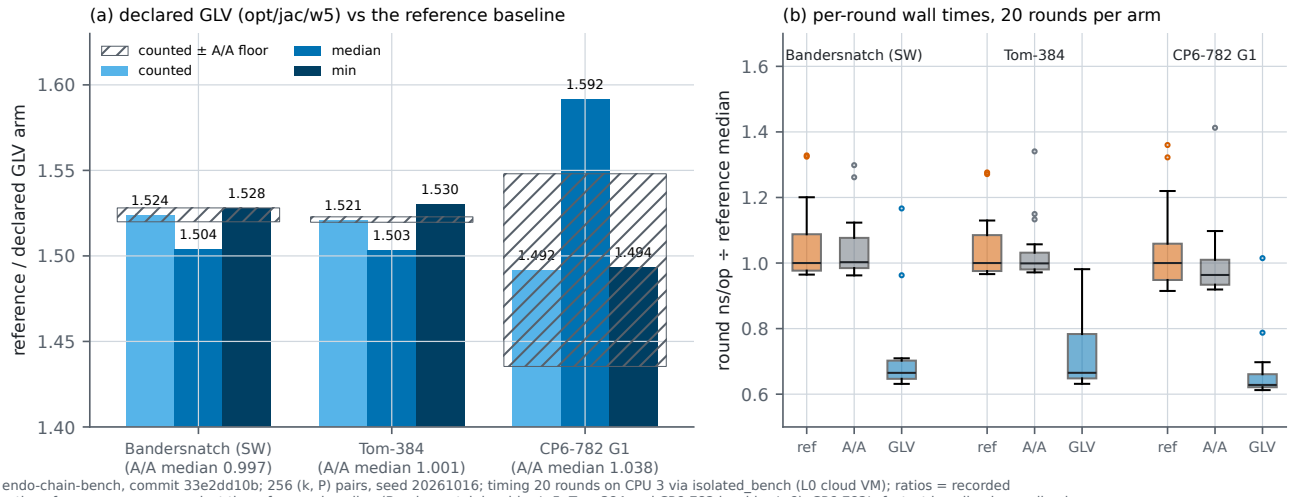
6 of 6 runs completed, each admitted at the first attempt. One earlier attempt to launch the counting runs under `/usr/bin/time` failed before the bench started because that tool is not installed; it produced no record.

3. Measured values



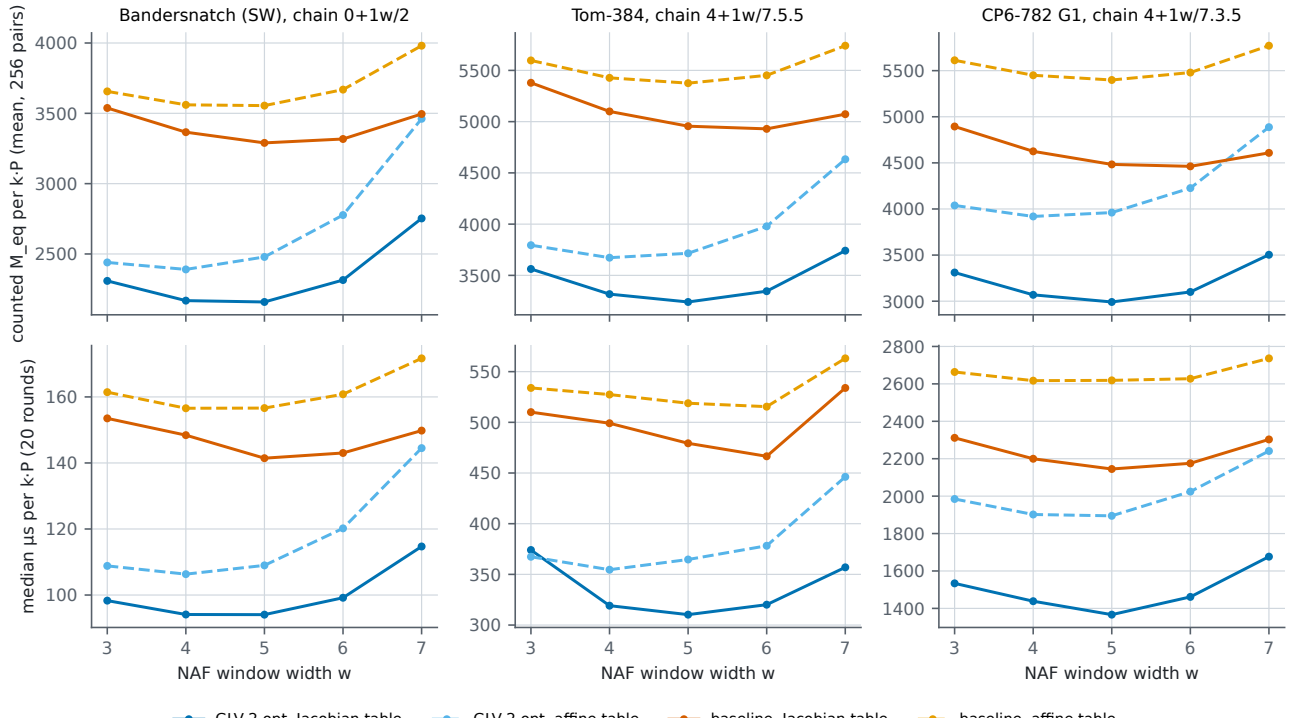
Counting runs (bandersnatch.counts.json, tom384.counts.json, cp6_782.counts.json), opcount build at commit 33e2dd10b; counted M_eq = M + S, mean over 256 pairs; model averages 400 other scalars

Figure 1: Counting runs. Unit: field operations $M_{\text{eq}} = M + S$ per $k \cdot P$, mean over 256 pairs. (a) Derived: min counted M_{eq} over baseline arms / min over GLV arms, and the same from the model column. (b) Measured deviation from the frozen model for every total arm.



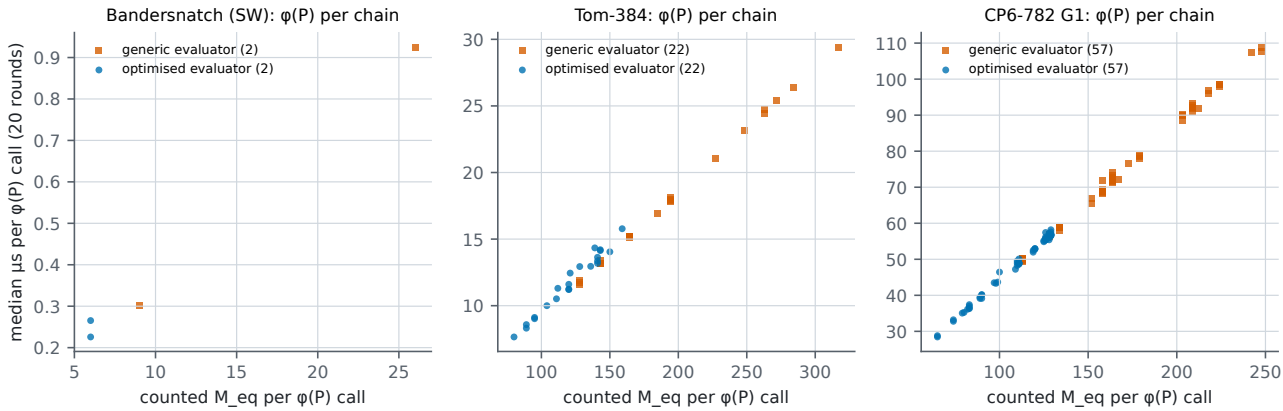
endo-chain-bench, commit 33e2dd10b; 256 (k, P) pairs, seed 20261016; timing 20 rounds on CPU 3 via isolated_bench (L0 cloud VM); ratios = recorded ratio_reference_over_arm against the reference baseline (Bandersnatch jacobian/w5, Tom-384 and CP6-782 jacobian/w6); CP6-782's fastest baseline by median is jacobian/w5 (README condition 4: 1.570)

Figure 2: Timing runs, n = 20 rounds per arm. (a) Recorded `ratio_reference_over_arm` of the declared GLV arm (counted, median, min); hatched band: counted ± $|1 - \mathrm{A/A\ median\ ratio}|$. (b) Per-round ns/op ÷ reference median.



endo-chain-bench, commit 33e2dd10b; 256 (k, P) pairs, seed 20261016; timing 20 rounds on CPU 3 via isolated_bench (L0 cloud VM); top: counting runs, bottom: timing runs

Figure 3: Grid of the declared chain: counted mean M_{eq} and median µs per $k \cdot P$ against width.



endo-chain-bench, commit 33e2dd10b; 256 (k, P) pairs, seed 20261016; timing 20 rounds on CPU 3 via isolated_bench (L0 cloud VM); stage diagnostics, not a speed of k·P; counts identical on every input (min = max)

Figure 4: $\varphi(P)$ stage diagnostics: counted M_{eq} per call (identical on every input) against median µs per call.

Recorded values for the declared GLV arm (README "Headline"): counted M_{eq} 2 158.6 / 3 240.4 / 2 991.5 against reference 3 289.8 / 4 929.6 / 4 462.4; median ns 94 059 / 310 273 / 1 366 596 against 141 443 / 466 490 / 2 175 424; A/A median ratios 0.997 / 1.001 / 1.038. Per-round spread (max/min − 1) on the reference 38 %, 32 %, 49 %; on the declared arm 85 %, 55 %, 66 %.

4. Raw records

`../../bandersnatch.counts.json`, `../../tom384.counts.json`, `../../cp6_782.counts.json`; `../../<curve>.timing.json`; `../../<curve>.isolated_bench_record.jsonl`; `.stdout.md` tables beside each; inputs in `../../frozen_inputs/` (`SHA256SUMS`).